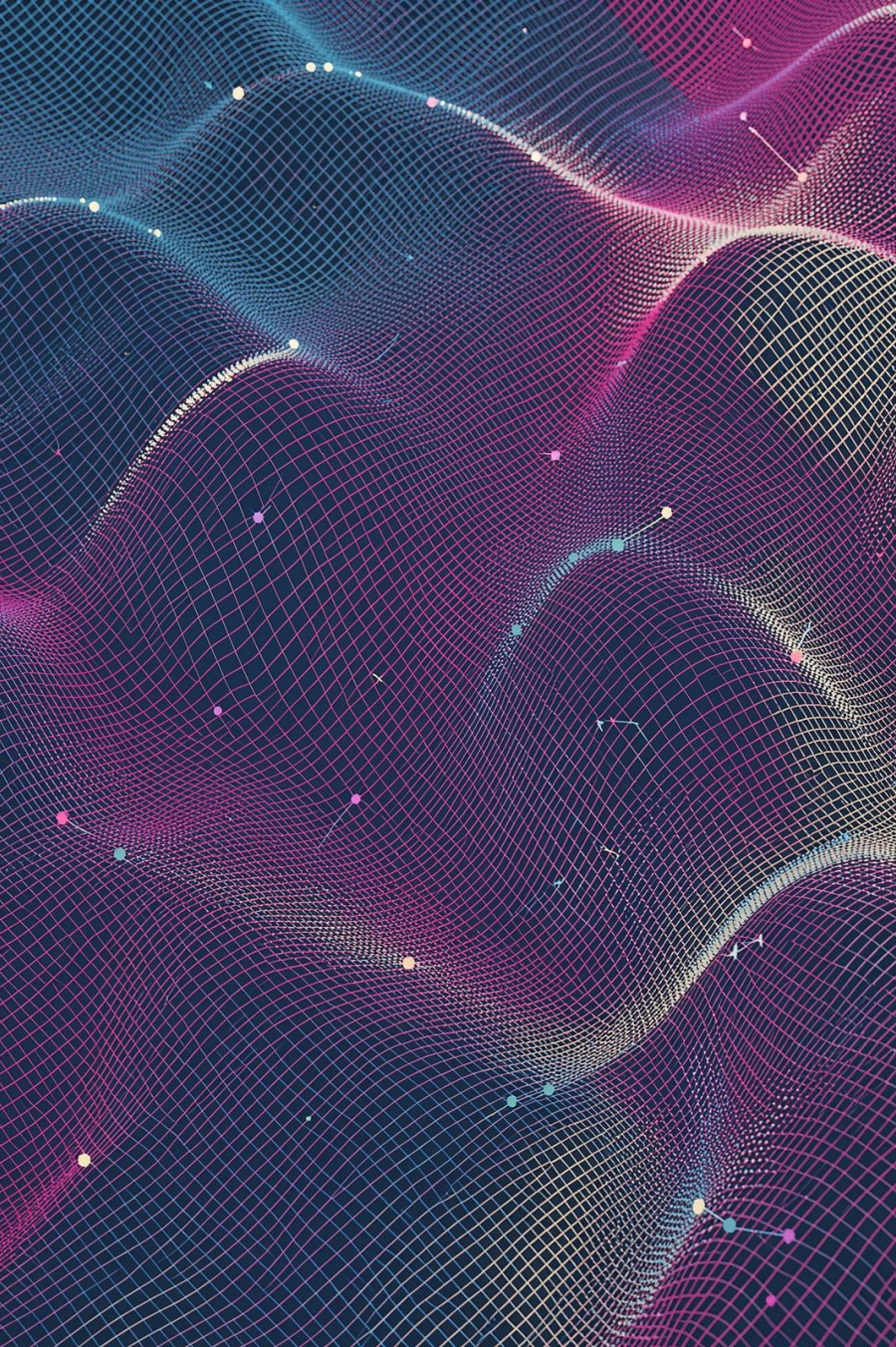




Oak & Ember AI Sales Assistant - Sales Genie

Production-Ready Product Recommendation Engine with Guardrails & Observability

Animesh Deb

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The Problem

Channel Chaos

Customer inquiries scattered across chat and email with no unified intake layer.

Slow & Inconsistent

Manual product matching creates bottlenecks and uneven customer experiences.

Manual Errors

Manual entry of names and emails into Excel increases chances of mistakes, duplicates, or omissions.

Hallucination Risk

LLM-generated responses can fabricate product claims, threatening brand credibility.

Inefficient Product Recommendations

Recommending the product manually based on their enquiry and budget takes lots of time and is error prone.

No Real time CRM logs

No standard lead scoring; CRM logging not real-time.

The Solution

Dual-Channel Intake

Streamlit chat UI +
Gmail automation —
one intelligent layer for
every customer
touchpoint.

Catalog Grounded Recommendations

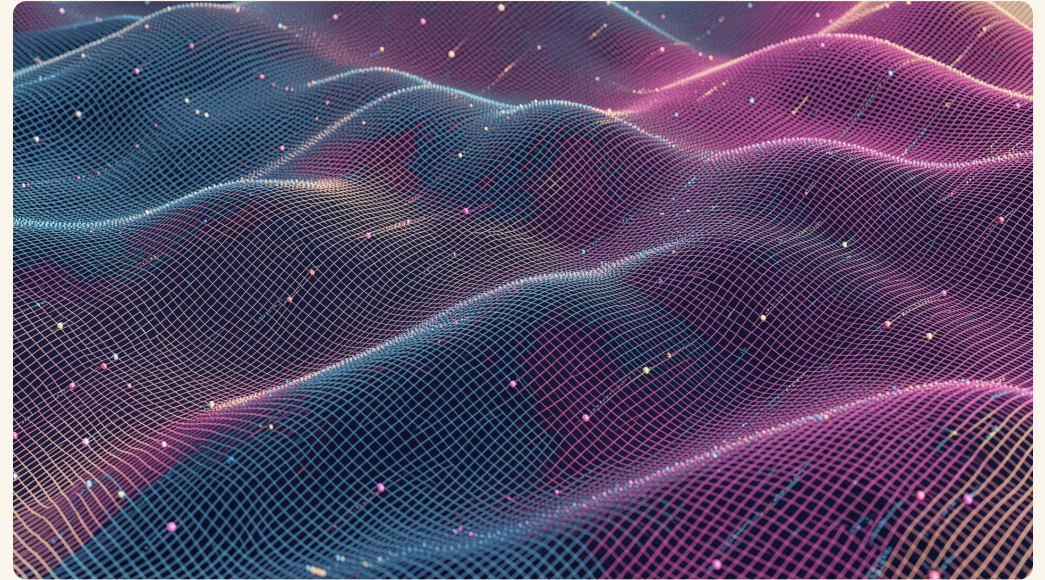
Only eligible and
products matching
customer requirements
are recommended.
Guardrails implemented
to filter off prompt
injections

Genuine Product Enquiries

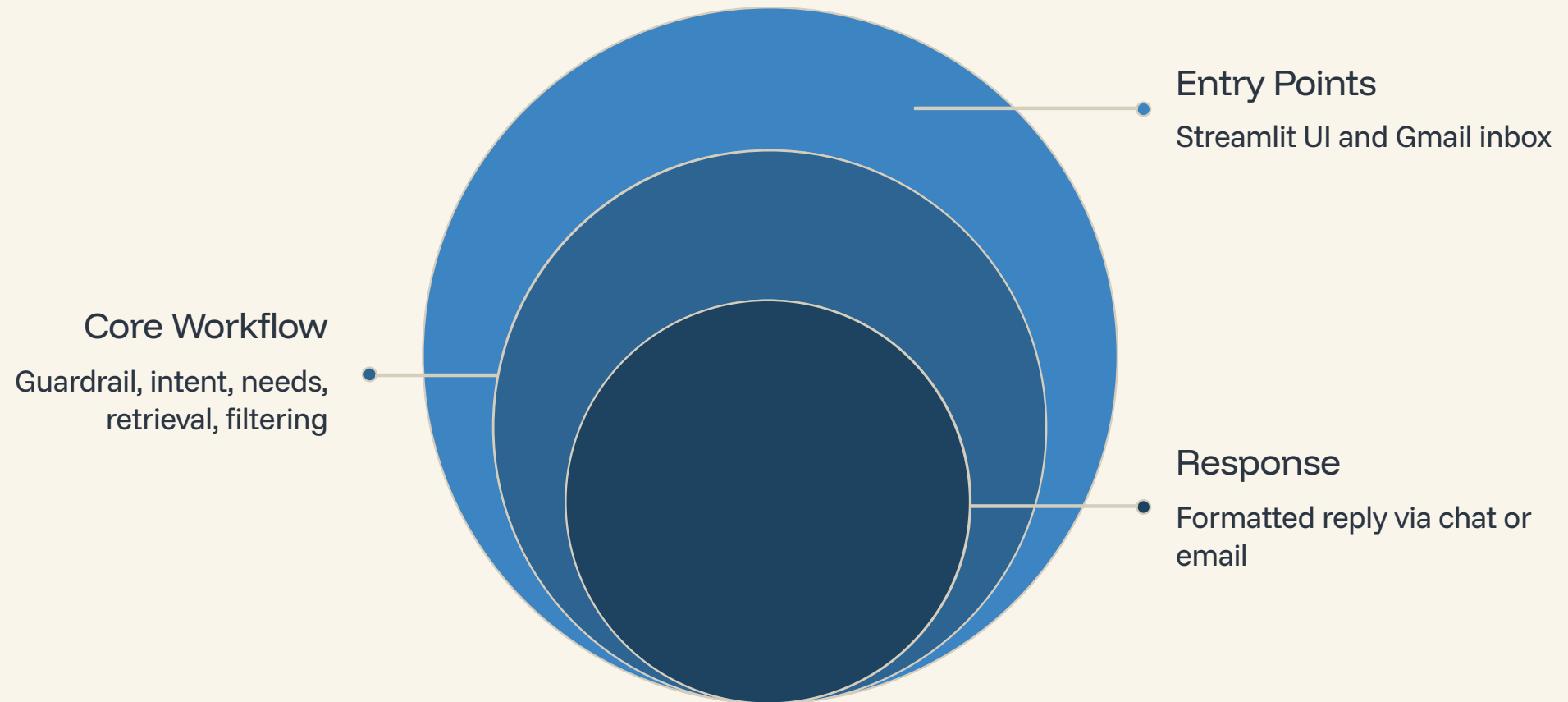
Auto filtering and
ranking of retrieved
products. Also cater
Unsubscription
requests

Real time CRM

Logging of real time
Customer data and
lead generation



System Architecture



Every customer request — regardless of channel — is processed through the same rigorous, multi-stage pipeline before a response is delivered.

Query-to-Response Flow

01

Guardrail

Detect prompt injection, unsafe requests, off-topic queries.

02

Intent & Needs

Extract customer requirements and budget constraints.

03

Retrieval

Vector search returns top 5 semantically relevant products.

04

Filter → Rank → Recommend

Apply hard constraints, score eligible products, LLM explains each fit.

05

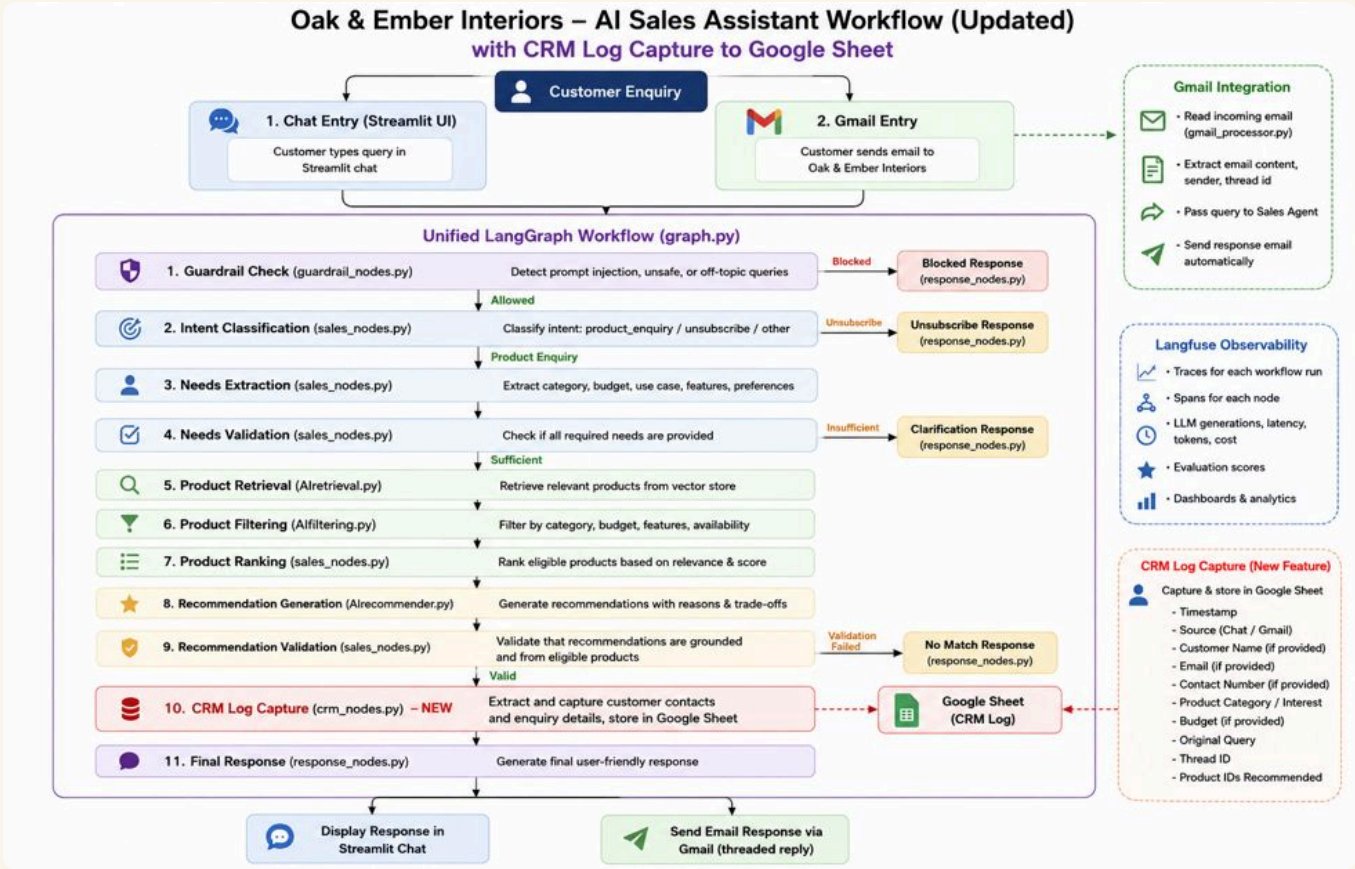
Validation

Verify product IDs, prices, and claims against live catalog before delivery.

06

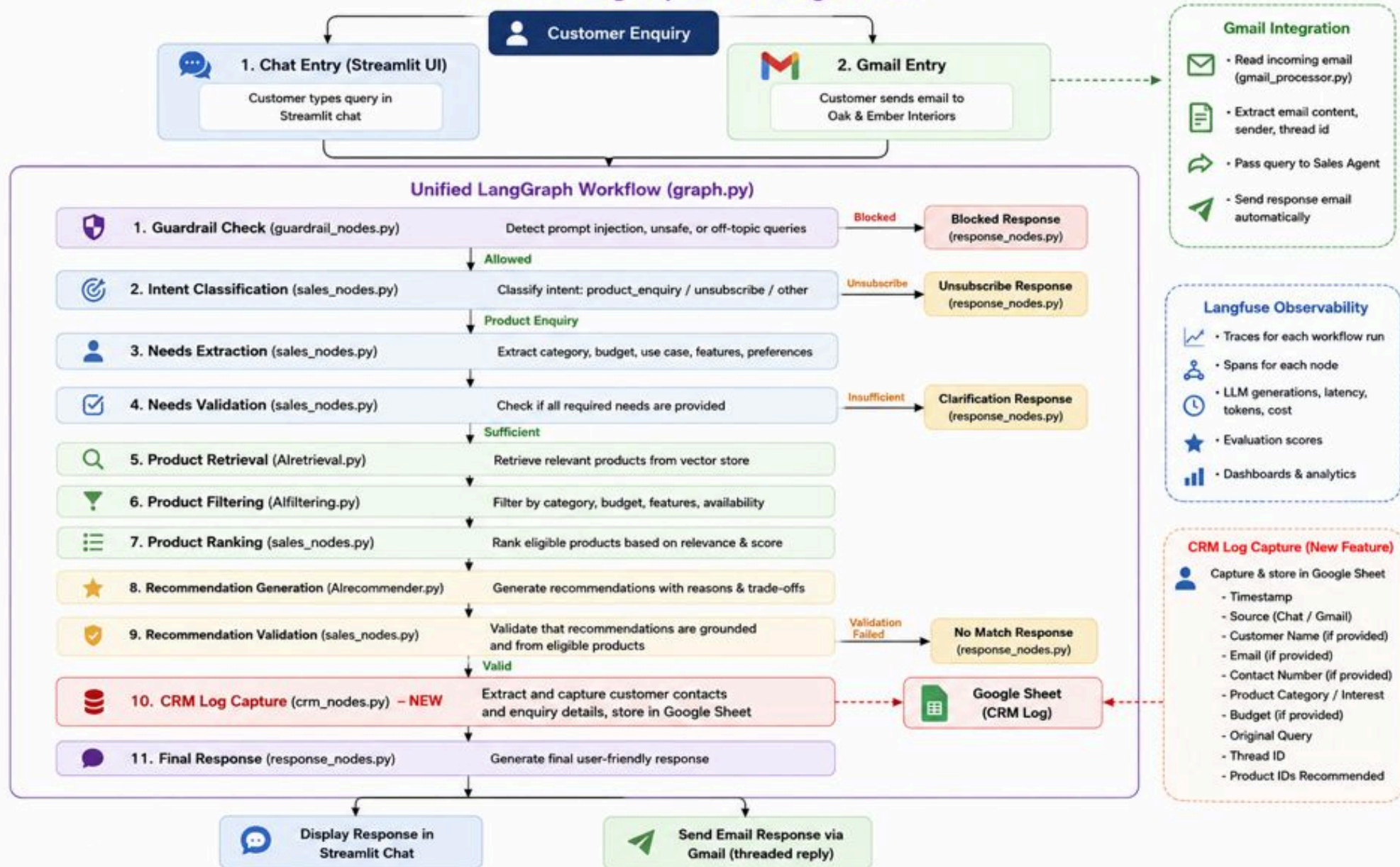
CRM Logs and Response

The product enquiry and customer data is logged and validated Response delivered to customer



Oak & Ember Interiors – AI Sales Assistant Workflow (Updated)

with CRM Log Capture to Google Sheet



Guardrails & Validation

Input Layer

Blocks prompt injection, flags unsafe requests, and rejects off-topic queries before any LLM call is made.

Recommendation Layer

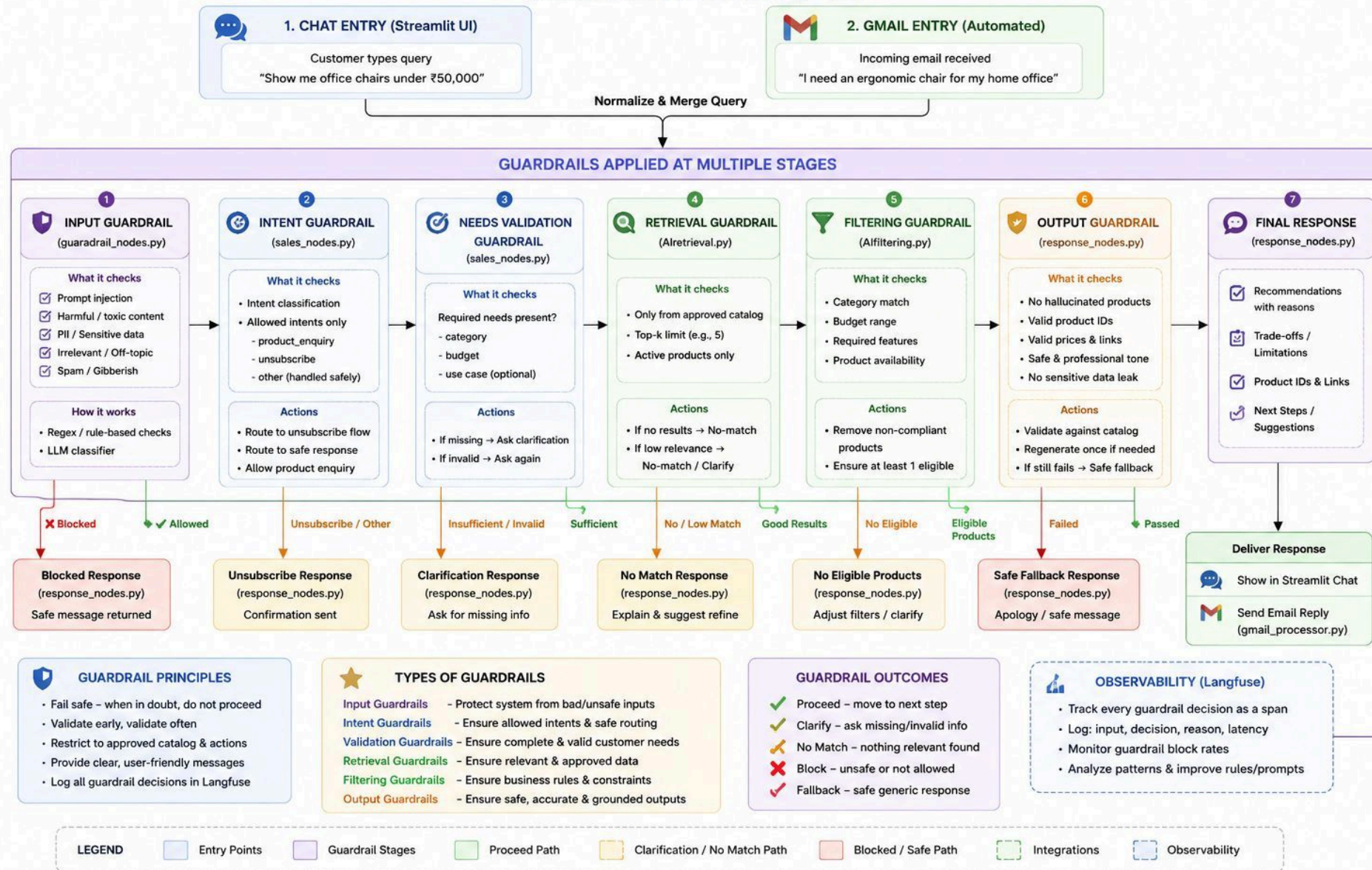
Hard constraints on budget and features enforced deterministically. All claims grounded exclusively in the product catalog.

Gmail Layer

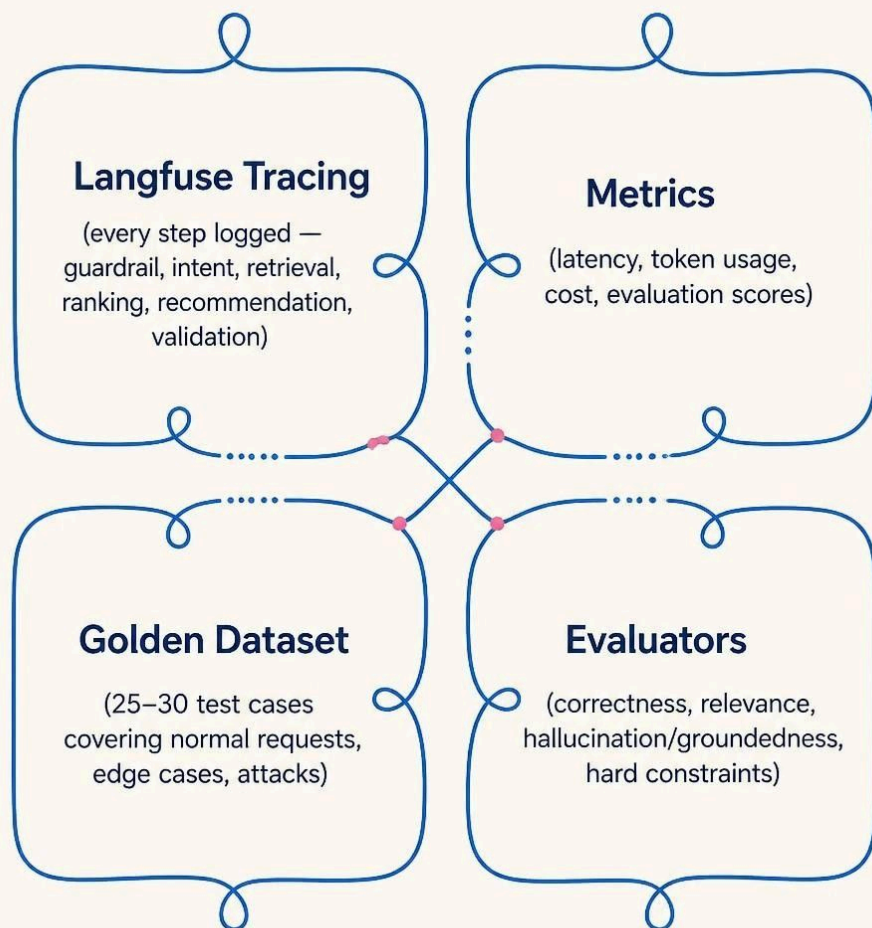
Duplicate prevention, correct sender routing, and zero false-success responses — reliable automation at every step.

Guardrail Architecture

Oak & Ember Interiors – AI Sales Assistant Guardrails Architecture & Flow



Observability & Evaluation



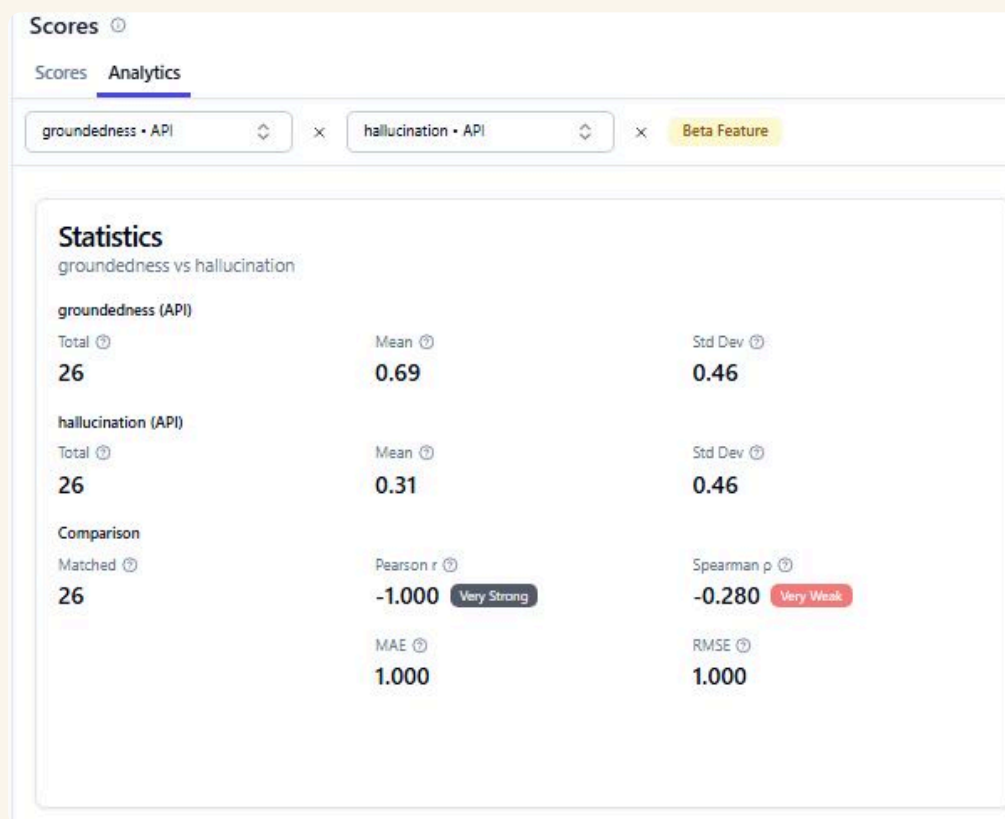
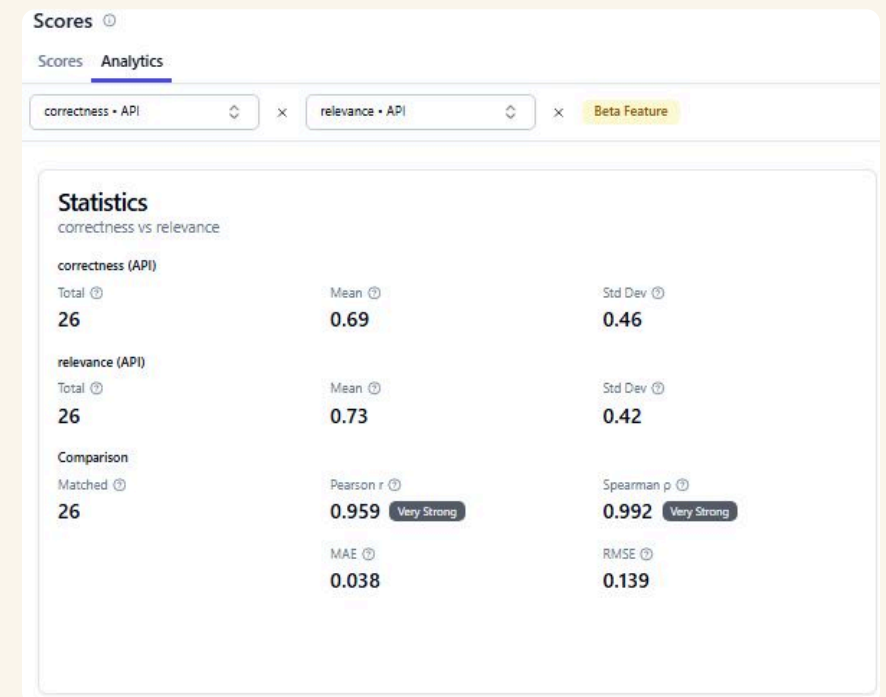
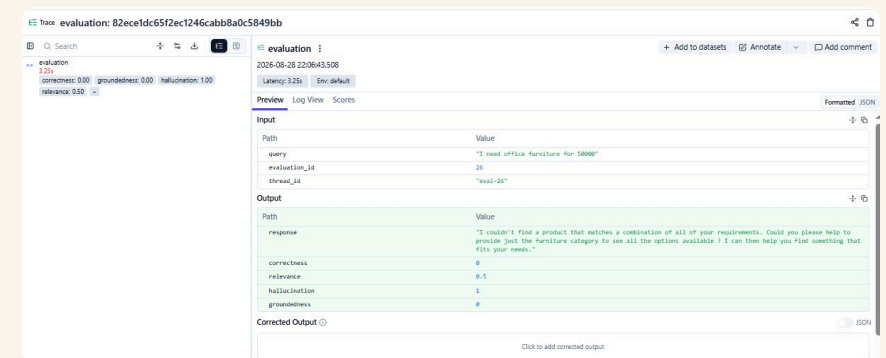
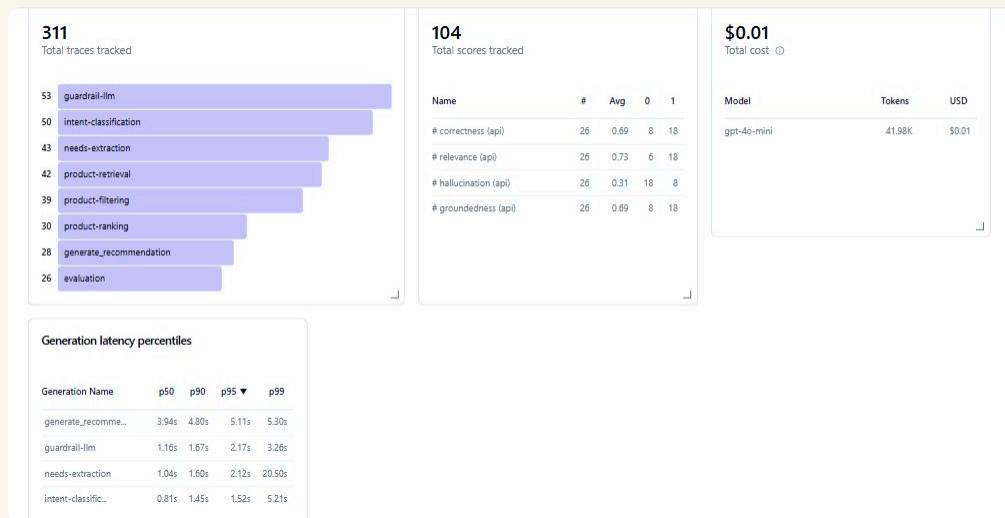
Full-Stack Observability

Every step of the pipeline is traced end-to-end with **Langfuse** — providing real-time visibility into latency, token cost, and evaluation scores.

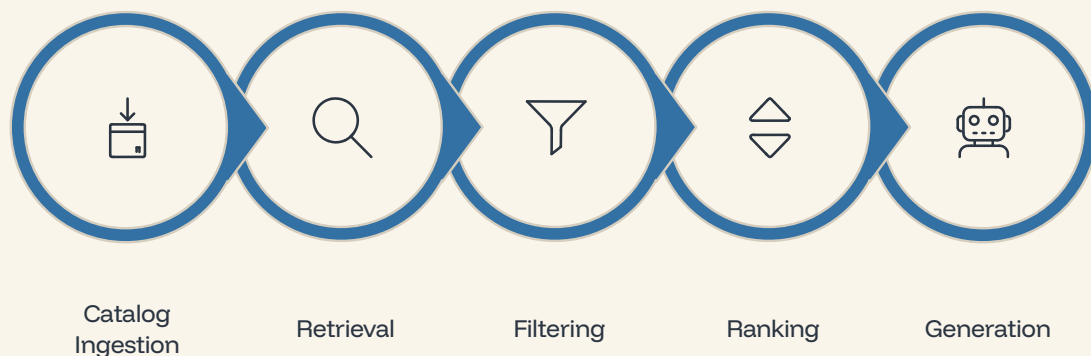
A curated **golden dataset** of 25–30 test cases stress-tests normal flows, adversarial attacks, and edge cases. LLM-as-judge evaluators measure correctness, relevance, groundedness, and constraint adherence automatically.

❏ Most AI projects skip structured evaluation. This one doesn't.

Observability and Evaluation Dashboard



RAG Pipeline



Why This Design Matters

Separating retrieval, filtering, and generation into discrete stages ensures the LLM never sees ineligible products — eliminating a primary source of hallucination at the architectural level.

Deterministic Python rules handle hard constraints; the LLM handles language. Each layer does only what it does best.

Technical Skills & Stack



AI / ML

LangGraph, LangChain, vector embeddings, LLM-as-judge evaluation



Integration

Gmail API, OAuth, fully automated email intake and response



Backend

Python, Pydantic, LangGraph state management



Observability

Langfuse tracing, structured logging, golden-dataset evaluation metrics

Streamlit UI

<<

Oak & Ember Interiors

Your AI furniture sales assistant.

Conversation

Session ID: 9e793e03-6bec-4f04-86a6-e061cb23371b

Clear Conversation

*** CONNECTING

Deploy

Oak & Ember Interiors

Intelligent Furniture Recommendations

Please provide your furniture requirements below, specify furniture category, budget, and any specific features you desire.

AI assistance: Recommendations are generated from the approved Oak & Ember product catalog. Please verify current price and availability.

Tell me what furniture you're looking for...